

gist-sdh-multifocal-resected-m1-k4n8

Preclinical recommendations — forward-looking horizon scan

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Preclinical recommendations

A forward-looking horizon scan of candidate drugs, compounds, and treatment strategies that are earlier than clinical development, or not yet developed, but plausibly target this patient's stated features. These are research directions, not enrollable options or treatment recommendations.

1. SSTR2-targeted peptide receptor radionuclide therapy (177Lu-DOTATATE), selected by 68Ga-DOTATATE PET (radioligand)

Target: SDH-deficient GIST (cell-surface SSTR2 expression) **Stage:** Early translational **Evidence strength:** Emerging

The only candidate with human dSDH-GIST evidence and an approved, PET-confirmable target, but the support is two case reports and there is nothing to treat while the patient is NED.

Of the six, this is the only candidate with evidence in the patient's own tumor type rather than a borrowed model: a metastatic SDHA-mutant GIST that responded to 177Lu-DOTATATE for more than a year, plus a second SDHB-deficient GIST that lit up on DOTATATE PET. The receptor is both the target and the imaging handle, so it can be confirmed before anyone commits, and the radioligand is already approved for neuroendocrine tumors. Two things keep it short of strong: the response data are two case reports, and the SSTR2 expression they rest on is documented only at case level, so it is a weaker, less reliable predictor of benefit than a systematically measured one. With the patient currently NED there is also no lesion to image or treat right now.

Mechanism. SDH-deficient tumors tend to put somatostatin receptor 2 on the cell surface, the same biology that makes paragangliomas DOTATATE-avid. 177Lu-DOTATATE binds SSTR2 and delivers beta-radiation into the cell, and the receptor doubles as the imaging handle on 68Ga-DOTATATE PET.

Translatability. Highest in the set: human dSDH-GIST case reports with strong SSTR2A staining and a durable response, though expression is case-level only and likely heterogeneous.

Developability. Closest to reach: an FDA-approved radioligand and existing PRRT centers make a PET-gated trial feasible. The catch: NED now, so no target to image.

Counter-productive MoA. Low (Radioligand kills via targeted radiation, so little pathway-level conflict; the real limits are dosimetry and an absent target when NED.)

Key risks.

- SSTR2 expression in dSDH-GIST is documented only in case reports, so it may be heterogeneous or simply absent in a given patient.
- The response evidence is two anecdotes, not a trial.
- Renal and marrow dosimetry cap how many cycles are safe.
- With the patient NED, there is no measurable, PET-confirmable target to act on today.

Open questions.

- Does dSDH-GIST express SSTR2 consistently enough to predict benefit, or only in a subset?

- If disease recurs, would DOTATATE-PET positivity reproduce the case-report responses?

Key references. doi:10.21037/tcr-21-131; pmid:29188154

2. Pharmacologic (high-dose IV) ascorbate (repurposed drug)

Target: SDH-deficient GIST (succinate-driven redox stress and a blocked TET demethylation axis) **Stage:** In vivo (animal) **Evidence strength:** Emerging

Real in-vivo antitumor effect and a cheap, repurposable drug, but every supporting study is in PPGL, and ascorbate does not always kill, so the cross-tumor leap is the whole bet.

The supporting work is genuine in-vivo efficacy: in SDHB-low pheochromocytoma and paraganglioma models, pharmacologic ascorbate caused oxidative DNA damage, shrank metastatic lesions, and extended survival. The lesion driving that effect (SDH loss, then succinate, then a redox and methylation derangement) is exactly what defines this tumor, so the vulnerability has a fair chance of carrying over even though the histology differs. It is off-patent and already infused at gram-level doses in oncology trials, just never in dSDH-GIST. What keeps it below the radioligand: the proof is cross-tumor, and in some settings ascorbate protects cells rather than killing them.

Mechanism. SDH loss drives succinate up, and succinate jams the alpha-ketoglutarate-dependent TET demethylases this tumor's biology runs through. Pharmacologic ascorbate can refuel TET activity by keeping its catalytic iron reduced, and at gram-level plasma concentrations it also acts as a pro-oxidant, generating extracellular H₂O₂ that SDH-deficient cells are poorly equipped to clear.

Translatability. Moderate: genuine in-vivo antitumor effect, but in SDHB-mutant PPGL models, never in a dSDH-GIST model, so the redox vulnerability is assumed to transport.

Developability. Easy to test, hard to fund: an off-patent drug already given IV in oncology trials, but nobody owns it, which is why high-dose ascorbate stalls.

Counter-productive MoA. Moderate (Direction of effect is not guaranteed; in some contexts ascorbate is cytoprotective, which would blunt the intended kill.)

Key risks.

- The data are in PPGL, not GIST; the redox vulnerability is assumed to transport but has not been shown in dSDH-GIST.
- High-dose IV vitamin C can cause hemolysis in G6PD deficiency and oxalate nephropathy.
- Ascorbate is not reliably cytotoxic; in some settings it protects cells instead.

Open questions.

- Does an authentic dSDH-GIST model show the same redox sensitivity seen in SDHB-low PPGL?
- What plasma level and schedule would be needed to reach a pro-oxidant effect in GIST?

Key references. pmid:32152203

3. Pyruvate carboxylase (PC) dependency as a synthetic-lethal target (no

clinical inhibitor yet) (conceptual strategy)

Target: SDH-deficient GIST (aspartate-pool dependency on pyruvate carboxylation) **Stage:** In vivo (animal)

Evidence strength: Moderate

The cleanest, most reproduced metabolic vulnerability in the set, and also the least drugged: a validated target with no molecule to give and no GIST-specific data yet.

This is the most thoroughly reproduced biology in the set: two independent groups showed that SDH-null cells stall or die when pyruvate carboxylation is blocked, confirmed in xenografts and traced directly by stable-isotope labeling. It is one of the cleaner metabolic synthetic-lethal nodes in SDH-loss disease. The reason it sits at rank 3 rather than higher is the gap to a patient. There is no pyruvate carboxylase inhibitor anyone could give, so this is a validated target, not a therapy, and the validation was done in kidney and PPGL cells rather than GIST.

Mechanism. Because the TCA cycle cannot run forward past complex II, the SDH-deficient cell drains its aspartate pool and refills it by carboxylating pyruvate to oxaloacetate through pyruvate carboxylase. That makes PC an enzyme they cannot do without: cut off pyruvate carboxylation and aspartate biosynthesis stalls.

Translatability. Low despite strong biology: the dependency is reproduced and in vivo, but in Sdh-null kidney and PPGL models rather than authentic dSDH-GIST.

Developability. Far from a patient: no selective PC inhibitor exists, and PC runs gluconeogenesis everywhere, so a tumor-selective window is unproven. This needs a discovery campaign.

Counter-productive MoA. Low (Tumors may reroute anaplerosis through glutamine, escaping PC blockade; otherwise little direct pathway conflict.)

Key risks.

- There is no molecule to give; this is target validation only.
- Systemic PC inhibition would risk hypoglycemia and broad metabolic toxicity.
- The dependency was shown in kidney and PPGL models, not GIST.
- Cells may switch to glutamine anaplerosis as a resistance route.

Open questions.

- Can a selective pyruvate carboxylase inhibitor or degrader be built with a real therapeutic window?
- Does the aspartate dependency hold in authentic dSDH-GIST, or do those cells escape through glutamine?

Key references. pmid:26302408; pmid:26522426

4. GRPR-targeted radioligand therapy (e.g. ¹⁷⁷Lu-NeoB / RM2), selected by ⁶⁸Ga-NeoB PET (radioligand)

Target: SDH-deficient GIST (cell-surface gastrin-releasing peptide receptor expression) **Stage:** Early translational **Evidence strength:** Emerging

A genuinely independent surface target for a drug-poor subtype, but the dSDH-GIST evidence is imaging only, with no therapeutic outcomes yet and no lesion to image while NED.

GRPR is a second surface target for this subtype, independent of SSTR2, which matters when a tumor has so few drugs. A 2025 study used 68Ga-NeoB PET to show GRPR expression across SDH-deficient wild-type GIST lesions and framed it as patient selection for future therapy trials. It ranks below the DOTATATE option for one plain reason: the GIST work stops at imaging. No GRPR radioligand has been given as therapy to a GIST patient, so there are no response data, only expression, and that expression looks heterogeneous, which makes it a weak predictor of benefit. The patient being NED removes any target to image today as well.

Mechanism. Gastrin-releasing peptide receptor sits on the surface of SDH-deficient wild-type GIST. A GRPR antagonist ligand such as NeoB carries a gallium-68 label for PET and a lutetium-177 label for therapy, so GRPR works as a paired imaging-and-treatment target the way SSTR2 does in neuroendocrine tumors.

Translatability. Lower than DOTATATE: in-vivo 68Ga-NeoB PET shows GRPR across dSDH-GIST lesions, but expression imaging only, with no therapeutic response data and likely heterogeneity.

Developability. Moderate: 177Lu-GRPR ligands (NeoB, RM2) are already in trials for other tumors, so the agent exists, but the dSDH-GIST work stops at imaging.

Counter-productive MoA. Low (Targeted radiation, so little pathway conflict; pancreatic physiologic GRPR uptake constrains dosimetry, and NED leaves no target.)

Key risks.

- Only expression and imaging data exist in GIST; there are no therapeutic outcomes to point to.
- GRPR levels are likely heterogeneous across lesions and patients.
- Physiologic pancreatic GRPR uptake constrains dosimetry.
- With the patient NED, there is no target to image or treat now.

Open questions.

- Does GRPR expression in dSDH-GIST translate into actual radioligand response?
- How does GRPR-directed PRRT compare with SSTR2-directed PRRT in the same patient?

Key references. doi:10.1186/s13550-025-01299-3

5. LDH (LDHA) inhibition with tool compounds (oxamate, GNE-140) (small molecule)

Target: SDH-deficient GIST (glycolytic NAD⁺ dependency from a broken TCA cycle) **Stage:** In vitro **Evidence strength:** Emerging

A clean in-vitro synthetic-lethal idea that follows directly from SDH biology, held back by murine-only data and the lack of any clinical-grade LDHA inhibitor.

When SDH is out, the cell leans on glycolysis, which forces a dependency on LDHA to regenerate NAD⁺. In an isogenic SDHC-loss model, both LDHA knockdown and the LDH inhibitors oxamate and GNE-140 cut viability preferentially in the SDH-deficient cells. The synthetic-lethal logic falls straight out of SDH biology. But this is an in-vitro signal in murine fibroblasts and chromaffin cells, a long way from gastric dSDH-GIST, and there is no selective, clinical-grade LDHA inhibitor, only academic tool compounds.

Mechanism. With SDH out, the TCA cycle is broken at complex II and the cell leans on glycolysis to make ATP, which forces a dependency on lactate dehydrogenase to regenerate NAD⁺ by converting pyruvate to lactate. Block LDHA and the SDH-deficient cell can no longer balance its redox books.

Translatability. Low: an in-vitro synthetic-lethal signal in murine SDHC-loss fibroblasts and chromaffin cells, never shown in vivo or in GIST, leaning on weak tool compounds.

Developability. Low: no drug-like, selective LDHA inhibitor has cleared tox and PK for humans. A real molecule would have to exist before this is testable.

Counter-productive MoA. Moderate (LDHA is broadly essential, so systemic inhibition risks normal-tissue toxicity; tumor-selective killing is unshown in vivo.)

Key risks.

- LDHA is broadly essential across normal tissues, and tumor-selective killing has not been shown in vivo for SDH-loss models.
- Tool compounds like oxamate (low potency) and GNE-140 carry off-target effects that muddy the synthetic-lethal claim.
- The models are murine fibroblasts and chromaffin cells, far from gastric dSDH-GIST.

Open questions.

- Can LDHA be inhibited selectively enough to spare normal tissue?
- Does the synthetic-lethal effect hold in an authentic dSDH-GIST model in vivo?

Key references. pmid:29464059

6. Ferroptosis induction (GPX4 or system xc- inhibition) (conceptual strategy)

Target: SDH-deficient GIST (iron and lipid-peroxidation redox vulnerability) **Stage:** Conceptual **Evidence strength:** Speculative

The most speculative entry, kept only as a direction to watch: a redox-stress hypothesis with no supporting experiment, no deliverable inhibitor, and a genuinely uncertain direction of effect.

This is a hypothesis, kept as a research direction rather than a result. The reasoning: SDH-deficient cells carry an iron and redox burden, and if losing complex II forces a harder reliance on GPX4 to hold off lipid peroxidation, then GPX4 or system xc- inhibition could push them into ferroptosis. The redox-stress angle has real support in SDH-loss models, the same weakness the ascorbate work exploits. But I could not find a single GPX4-inhibitor experiment in dSDH-GIST or even in dSDH-PPGL, so the specific ferroptosis claim is unproven, and the direction of effect could genuinely cut either way.

Mechanism. SDH-deficient cells carry an iron and redox burden, and complex II normally helps keep the lipophilic CoQ antioxidant pool reduced. If losing SDH forces a harder reliance on GPX4 and glutathione to hold off membrane lipid peroxidation, then inhibiting GPX4 or the cystine importer system xc- could push these cells into ferroptosis.

Translatability. Lowest: no supporting experiment in any SDH context, and direction of effect is uncertain, since intact SDH can support ferroptosis resistance in some settings.

Developability. Lowest: clinical-grade GPX4 inhibitors do not exist, only unstable tool compounds like RSL3. The one repurposable handle is weak system xc- inhibition via sulfasalazine.

Counter-productive MoA. Moderate (Direction of effect is genuinely uncertain; SDH loss may make cells more ferroptosis-resistant, so induction could fail.)

Key risks.

- Entirely unvalidated in any SDH context.
- Direction of effect is genuinely uncertain; in some settings intact SDH supports ferroptosis resistance, so SDH loss could cut either way.
- No deliverable GPX4 inhibitor exists.

Open questions.

- Does SDH loss make these cells more or less vulnerable to ferroptosis?
- Is there any SDH-deficient model where GPX4 inhibition has been tested at all?